Original Article

Cancer Incidence Rate Variations among the Chinese, South Asian, and Vietnamese in Massachusetts, 2011–2015

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Abstract: Objectives: To examine cancer incidence among the 3 Asian, non-Hispanic ethnicities with the highest frequency of cases (South Asian, Chinese, and Vietnamese). Methods: Age-adjusted incidence rates for all invasive cancers were calculated for South Asian (Indian, Pakistani), Chinese, and Vietnamese cancer cases reported to the Massachusetts Cancer Registry (MCR). Additionally, rates were calculated for the most frequent cancers among non-Hispanic Asians (prostate, colorectal, female breast, female thyroid, lung, and male liver). The 95% confidence intervals were calculated to determine statistical significance between the rates. Results: South Asian and Vietnamese females had significantly elevated rates of all invasive cancers compared to Chinese females, while Chinese and South Asian females had a significantly elevated breast cancer rate. Vietnamese males had a significantly elevated rate of all invasive cancers, liver cancer, and lung cancer compared to the other 2 groups. Due to the high rates of lung cancer among Vietnamese males, MCR current/previous smoking data were compared for all cancers. Among Vietnamese, Chinese, and South Asian male cancer cases, current/previous smoking percentages were 64%, 51%, and 35%, respectively. Conclusions: Our analyses showed a significant difference of rates for several cancers by specific Asian ethnicity within the broader Asian, non-Hispanic race category. Differences in tobacco use, maternal hepatitis B infection, and diet likely contribute to some of the differences. These data can aid in the development of prevention programs, such as smoking cessation and mammography screening that are culturally and linguistically specific within this large and diverse group.

Key words: *Asian ethnicity, Chinese peoples, South Asian peoples, Vietnamese peoples*

Background

According to the US Census definitions, *Asians* are people having origins in any of the original peoples of the Far East, Southeast Asia, or the Indian subcontinent.[1] Despite being part of the Asian continent, Arab people from the Middle East are, for the most part, counted as White non-Hispanics, as are Arabs from North Africa. According to the 2011–2015 American Community Survey of the US Census (ACS) estimates, Asian non-Hispanics constituted 5.9% of the general population in Massachusetts and 5.1% of the US population. The 2011–2015 ACS estimated that the Massachusetts Asian population was primarily Chinese (37%), South Asian (22%), Vietnamese (12%), Cambodian (8%), Korean (7%), Filipino (3%), and Japanese (2%).

Asians are routinely presented as 1 of the 4 main racial ethnic categories for cancer statistics released by the Massachusetts Cancer Registry (MCR). They are presented as Asian, non-Hispanic, though Asian Hispanics represented only 0.3% of the total Asian cases in the study period. In the MCR annual report for 2011–2015, non-Hispanic Asian males had the significantly lowest rate of all male cancer types combined with 312.4 (95% CI, 298.6–326.2) cases per 100,000 for the years 2011–2015 compared to 498.2 (95% CI, 494.5–501.8) for White non-Hispanics, 514.6 (95% CI, 498.4–530.7) for Black non-Hispanics, and 376.6 (95% CI, 362.0–391.1) for Hispanics. Non-Hispanic Asian females had the lowest rate, as well (307.2; 95% CI, 295.3–319.1), significantly lower than White non-Hispanics (464.5; 95% CI, 461.2–467.8) and Black non-Hispanics (399.1; 95% CI, 387.3–410.9), though not significantly lower than Hispanics (324.1; 95% CI, 315.9–338.0). Both non-Hispanic Asian males and females had lower rates for most individual cancers when compared to White and Black non-Hispanics and comparable rates to Hispanics, with the exception of liver cancer for both sexes.[2]

As described earlier, Asians are a very large and diverse group of people coming from an area thousands of miles from Japan to Pakistan and Mongolia to Indonesia. The language, diet, social norms, traditional healing practices, and religion of someone from India are very different than someone from China or Japan or the Philippines, yet they are grouped together under the auspices of Asian. A recent paper on Asian Americans and prostate cancer highlighted the differences among various Asian groups and showed the need for disaggregation of these data.[3] In order to better understand the diverse ethnic groups within the Asian category, this study examines in more detail the differences in cancer incidence between the 3 most populous Asian ethnic groups in Massachusetts: Chinese, South Asians (Indian and Pakistani), and Vietnamese.

Table 1 presents some demographic differences between South Asians, Chinese, and Vietnamese residents of Massachusetts.[4] The median income education levels are much higher among South Asians. The percentage of those American born is about equal among the 3 groups, while there are higher percentages of English speakers among South Asians. English is an official language in both Pakistan and India.

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Table 1. Characteristics of South Asian, Chinese, and Vietnamese People in Massachusetts, 2010–2014			
	South Asian	*Chinese*	*Vietnamese*
Nativity			
Native born	30.9%	31.5%	32.5%
Naturalized	28.7%	36.0%	49.4%
Not a citizen	40.4%	32.5%	18.1%
Language			
English only	22.0%	17.7%	10.2%
English spoken well	56.7%	29.5%	38.6%
English spoken not well	21.3%	52.8%	61.2%
Poverty Level			
Individual poverty rate	6.1%	16.0%	16.7%
Family poverty rate	3.5%	9.9%	15.6%
Income			
Median household income	$113,566	$69,581	$56,895
Education			
Graduate degree	52.6%	34.2%	6.7%
Bachelor's degree	29.7%	23.1%	19.2%
High school diploma	11.5%	25.0%	43.0%
No high school diploma	6.2%	17.7%	31.1%

Source: 2010-2014 American Community Survey.

Methods

Incident invasive cancers in Massachusetts residents from 2011–2015 were provided by the MCR. Approximately 9% of non-Hispanic Asians were reported to the MCR from 2011–2015 with a specific ethnicity not otherwise specified (NOS). Since having a large number of NOS cases would lead to an underestimate of incidence rates for these ethnic groups, we recoded these cases based on the distribution of selected cancer types among cases with a specified ethnicity. For example, among Asian non-Hispanic males with a specified ethnicity, 48% of prostate cases were Chinese, 16% were South Asian (Indian or Pakistani), 9% were Vietnamese, and 27% were other. Prostate cancer cases among Asian NOS males were then recoded based on these distributions.

Once the NOS cases were recoded, analyses were limited to cancers with the highest frequencies among non-Hispanic Asians (all invasive, female breast, colorectal, liver, lung, thyroid, and prostate) in order to obtain statistically viable results. Even with this, rates were not calculated for cancers with fewer than 20 cases. Although the methodology for recoding Asian, NOS yielded estimated numbers for these groups, it is not a replacement for having actual numbers. Still, the importance of using the available data on Chinese, South Asians, and Vietnamese for these estimations in order to highlight the differences in incidence rates within the Asian, non-Hispanic category outweighed the limitations placed on the data analysis by the data quality.

While the population data for the Asian, non-Hispanic group as a whole and the total Massachusetts group came from US intercensal figures, the population estimates for the Chinese, Vietnamese, and South Asians were from the US Census 2011–2015 American Community Survey (ACS) estimates. As with the NOS reclassification methods, there are limitations to ACS data in terms of precision of the data resulting from smaller sample sizes. Incidence and mortality rates are per 100,000 and are age-standardized to the 2000 United States Census population. All statistical analyses were performed using SAS 9.3 (SAS Institute Inc).[5] We calculated 95% confidence intervals for incidence rates for the period and then examined the intervals for overlap to determine statistically significant differences (no overlap) or no significant differences (overlap). Rates were compared between South Asians, Chinese, and Vietnamese. These groups were then compared to all Asians and to all invasive cancer cases in Massachusetts, regardless of race/ethnicity.

Results

South Asian and Vietnamese females had significantly elevated rates of all invasive cancers compared to Chinese females, though significantly lower than all Massachusetts females. Additionally, the incidence rate for Vietnamese females was significantly elevated compared to all Asians. South Asian females had a breast cancer rate that was significantly elevated compared to all other Asian groups, but comparable to all Massachusetts females. Colorectal and lung cancer rates did not differ significantly within the Asian groups though only the rates among South Asian and Vietnamese females were significantly lower than all Massachusetts females. There were no significant differences in thyroid cancer between any of the groups, including all Massachusetts females (Table 2).

Vietnamese males had a significantly elevated rate of all invasive cancers compared to the other Asian groups, but significantly lower than all Massachusetts males. Like Vietnamese females, the incidence rates for males was significantly elevated compared to all Asians. While prostate and colorectal cancer rates were significantly lower than all Massachusetts males and comparable to all Asians, Chinese males did have a significantly elevated rate of prostate cancer compared to Vietnamese males. Compared to South Asian males, Chinese and Vietnamese males had significantly elevated rates of lung cancer. Vietnamese males in turn had a significantly elevated lung cancer rate compared to Chinese males. Vietnamese males also had a significantly elevated lung cancer rate compared to all Asian and Massachusetts males. The same was true for liver cancer with Vietnamese males having a significantly elevated rate compared to the other Asian groups and all Massachusetts males (Table 3).

Due to the high rates of lung cancer among Vietnamese males, tobacco use data among all cancers in the MCR were analyzed. Among Vietnamese, Chinese, and South Asian males, where there were significant lung cancer differences, the percentages of current or previous tobacco use were 64% (Vietnamese), 51% (Chinese), and 35% (South Asian). These percentages are likely an underestimate.[6]

Table 2. Age-Adjusted Cancer Incidence Rates (95% CI) among Specific Asian Ethnicities Compared to All Asians and All Massachusetts* Cases, Females, 2011–2015					
Cancer	*South Asian (n = 484)*	*Chinese (n = 1,159)*	*Vietnamese (n = 324)*	*All Asian (n = 2,226)*	*All Massachusetts (n = 95,757)*
All invasive	324.9 (296.0–353.9)	270.1 (254.5–285.6)	369.7 (329.5–410.0)	309.1 (297.6–320.6)	450.9 (448.0–453.8)
Breast	126.5 (108.7–144.4)	77.2 (69.1–85.3)	83.3 (67.1–99.6)	91.1 (85.1–97.2)	137.6 (136.0–139.3)
Colorectal	20.4 (12.7–28.1)	27.6 (22.5–32.7)	22.0 (13.4–30.7)	28.7 (25.0–32.3)	33 (32.5–33.9)
Lung	NA	33.2 (27.5–38.8)	48.5 (29.8–67.1)	33.1 (29.1–37.2)	60.2 (59.1–61.2)
Thyroid	32.4 (24.4–40.5)	27.4 (22.6–32.2)	37.7 (27.2–48.1)	29.7 (26.4–33.0)	29.6 (28.7–30.4)

* *All Massachusetts* indicates all cancer cases in Massachusetts regardless of race/ethnicity. Rates are per 100,000 and were age adjusted to the 2000 US Standard Population. *NA* indicates fewer than 20 cases.

Table 3. Age-Adjusted Cancer Incidence Rates (95% CI) among Specific Asian Ethnicities Compared to All Asians and All Massachusetts* Cases, Males, 2011–2015					
Cancer	*South Asian (n = 373)*	*Chinese (n = 1,000)*	*Vietnamese (n = 348)*	*All Asian (n = 2,777)*	*All Massachusetts (n = 87,884)*
All invasive	261.6 (235.0–288.1)	314.0 (294.6–333.5)	412.5 (369.2–455.8)	313.9 (300.8–326.9)	493.9 (490.6–497.3)
Prostate	62.6 (48.5–76.7)	68.5 (59.4–77.6)	42.9 (29.6–56.3)	57.3 (51.8–63.2)	106.3 (104.8–107.8)
Colorectal	20.6 (13.0–28.3)	32.8 (26.6–39.0)	52.4 (36.5–68.2)	32.4 (28.2–36.6)	41.8 (40.9–42.8)
Lung	32.3 (22.2–42.4)	62.3 (53.5–71.2)	106.4 (83.5–129.3)	57.9 (51.9–63.8)	69.3 (68.0–70.6)
Liver	NA	22.3 (17.3–27.3)	70.9 (53.2–88.5)	26.1 (22.4–29.8)	12.9 (12.4–13.4)

* *All Massachusetts* indicates all cancer cases in Massachusetts regardless of race/ethnicity. Rates are per 100,000 and were age adjusted to the 2000 US Standard Population. *NA* indicates fewer than 20 cases.

Conclusions

Our analyses showed a significant difference of rates for several cancers by specific Asian ethnicity within the broader Asian, non-Hispanic race category. The significantly lower rates for prostate and colorectal cancers among all Asian ethnicities compared to total rates found in this study has been previously described.[2] In the current study, there were some variations within Asian ethnic groups with Vietnamese males having a significantly lower prostate cancer incidence rate compared to Chinese males. Both groups, however, did not differ significantly from the prostate and colorectal rates for all non-Hispanic Asian males.

Breast cancer rates were highest among South Asian females, higher than what was reported nationally for 2008–2012.[7] Those national figures showed South Asian and Chinese with comparable breast cancer rates, but higher than Vietnamese females and lower than non-Hispanic white females. The current study found breast cancer rates among South Asian females to be significantly higher than the rate for all non-Hispanic Asian females and comparable to all Massachusetts cases. A 2010 study on breast cancer screening among South Asians suggested that the adoption of a western lifestyle may increase the risk of breast cancer among Asian immigrants.[8]

Vietnamese males had significantly elevated incidence rates of lung and liver cancer compared to all non-Hispanic Asian males and all Massachusetts males. These 2 cancers are the most common cancers in Vietnam among both males and females. The Vietnamese government has put tobacco control policies into place, resulting in a drop from 56% of all males smoking to 47% from 2002–2010.[9] Additionally, most newborns in Vietnam are immunized against hepatitis B to prevent future infections and cases of liver cancer.[9]

This study highlighted some of the ethnic differences in cancer incidence within the category of Asian, non-Hispanic in the Massachusetts Cancer Registry. While most of the cancers among non-Hispanic Asian ethnicities examined for this study did not differ significantly from all Asian, there were some notable differences in female breast cancer, lung, and liver cancer. These Asian groups differ not only in cancer rates, but also exhibit significant differences in income, education, and language, all of which can limit access to health care. Additionally, smoking data prevalence among adults (aged ≥18 years) from the 2016 National Survey on Drug Use and Health showed Vietnamese having higher percentages (16.3%) compared to South Asians and Chinese (both 7.6%).[10] The knowledge of all these differences can aid in the development of prevention programs, such as smoking cessation, mammography, and hepatitis B vaccination, that are culturally and linguistically specific within this large and diverse group.

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